

GOVERNMENT OF JHARKHAND • DEPARTMENT OF HIGHER & TECHNICAL EDUCATION

Product Requirements Document (PRD)

Societal Innovation Collaboration Platform

SIH Problem Statement 26043 — Software / Disaster & Societal Management

DOCUMENT VERSION

v1.0 (Production / Prototype MVP)

DATE OF CREATION

August 2026

TARGET ORGANIZATION

Dept of Higher & Technical Education,
Jharkhand

CLASSIFICATION

Official Hackathon Prototype Specification

CORE TECHNOLOGIES

FastAPI, React.js, Supabase PostgreSQL,
Hugging Face AI

NEP ALIGNMENT

NEP 2020 Experiential & Community Innovation

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1. Executive Summary

The **Societal Innovation Collaboration Platform** is an end-to-end, AI-powered digital ecosystem built specifically to solve the systemic disconnect between grassroots citizen problems in Jharkhand and the academic, research, and technical capabilities of Higher Education Institutions (HEIs) and industry leaders.

Currently, citizens across urban and rural Jharkhand identify thousands of urgent localized challenges in education, public healthcare, clean drinking water, sanitation, agricultural management, and disaster resilience. However, there is no structured digital channel to submit, auto-classify, route, and track these challenges to completion. Simultaneously, universities possess talent and research mandates under **National Education Policy (NEP) 2020** but lack direct exposure to validated real-world problems.

CORE VALUE PROPOSITION

This platform acts as an intelligent bridge: transforming crowdsourced community challenges into active university research projects, student capstone innovations, and corporate social responsibility (CSR) funded industry ventures with full lifecycle transparency and real-time visual analytics.

2. Project Overview & SIH 26043 Context

Parameter	Specification Details
Problem Statement ID	26043
Problem Statement Title	A digital platform to crowdsource societal challenges and facilitate collaborative problem solving through universities and industry partnerships
Category	Software Development (JAMstack + AI Integration)
Theme	Disaster Management & Societal Welfare
Nodal Organization	Department of Higher & Technical Education, Government of Jharkhand
Target Scope	Statewide deployment across 24 districts, 50+ HEIs, and 500+ student research teams

3. Problem Statement & Gap Analysis

3.1 Current Challenges in Jharkhand

Communities across Jharkhand regularly encounter critical domain-specific challenges:

- **Water & Sanitation:** Groundwater contamination, lack of filtration in rural schools, open drainage issues.
- **Agriculture:** Soil acidity, irregular rainfall monitoring, lack of local cold chain technology.
- **Healthcare:** Remote telemedicine gaps, maternal health monitoring in tribal belts.
- **Education & Skill:** Digital literacy divide, vocational lab infrastructure constraints.

3.2 Key Systemic Gaps Identified

Identified Gap	Impact on Ecosystem	Platform Solution
No Central Channel	Citizens cannot report issues to academic experts.	Mobile-responsive crowdsourcing form with GPS & media.
Manual Triage Delay	Manual routing of problems takes weeks or months.	Zero-Shot NLP AI classification & auto-routing in <2 sec.
Unrealized NEP 2020	Students work on synthetic/theoretical projects.	Real-world demand-driven problem assignments.
Siloed Industry CSR	Corporates fail to discover viable social projects.	Dedicated Industry Partnership & Funding portal.

4. Vision, Mission & Core Strategic Objectives

4.1 Vision

To establish a transparent, scalable, and AI-driven innovation ecosystem that transforms grassroots community challenges into practical research, sustainable startups, and measurable social impact across Jharkhand.

4.2 Mission

To empower citizens, university researchers, student teams, government officials, and industry sponsors through a unified digital platform that automates problem identification, intelligent matching, milestone tracking, and solution deployment.

4.3 Key Performance Targets

Strategic Objective	Year 1 Target	Year 2 Target
Crowdsourced Submissions	1,000+ Verified Challenges	10,000+ Verified Challenges
University Onboarding	50+ Higher Education Institutions	150+ Technical Institutions
AI Categorization Accuracy	90%+ Zero-Shot Accuracy	95%+ Fine-Tuned NLP Accuracy
Project Resolution Rate	60% Active Lifecycle Progress	85% Completed & Deployed Solutions

5. User Personas

Persona 1: Citizen Reporter (Grassroots User)

Name: Suresh Kumar (Ranchi District) | **Role:** Agriculturalist & Community Leader

- Needs:** Simple UI, regional language support, voice/photo submission, location tagging.
- Pain Point:** Complaints sent to district offices are delayed or lost without tracking numbers.
- Platform Workflow:** Submits issue via mobile web with photo and GPS location; receives SMS tracking updates.

Persona 2: University Administrator / Dean of Research

Name: Dr. Priya Sharma | **Role:** Dean of R&D, Central University of Jharkhand

- **Needs:** Automated queue of domain-relevant challenges, team assignment tools, milestone validation.
- **Pain Point:** Difficulty finding validated real-world problems for final-year engineering students.
- **Platform Workflow:** Accepts auto-routed "Water Management" challenges and assigns student project teams.

Persona 3: Student Project Lead

Name: Aishwarya Patel | **Role:** Final Year B.Tech Computer Science Student

- **Needs:** Task board, repository upload, mentor guidance, clear project milestones.
- **Platform Workflow:** Submits project proposals, updates status from *Assigned* → *In Progress* → *Deployed*.

Persona 4: Industry CSR & Funding Sponsor

Name: Mr. Ravi Desai | **Role:** Head of CSR Innovation, Tata Steel

- **Needs:** Project discovery catalog, verifiable social impact metrics, IP/startup co-incubation.
- **Platform Workflow:** Filters high-impact agriculture prototypes and allocates CSR grant funding.

Persona 5: Government Nodal Official

Name: Mr. Aman Singh | **Role:** Secretary, Dept. of Higher & Technical Education

- **Needs:** High-level visual dashboard, district heatmaps, university leaderboard, analytics export.
- **Platform Workflow:** Monitors district-wise challenge resolution rates for executive decision-making.

6. Functional Requirements

Module 1: Citizen Engagement Module (Web & Mobile)

Req ID	Requirement Description	Priority	Acceptance Criteria
CE-1.1	Responsive submission interface across mobile and desktop.	P0	100% responsive layout, <3s mobile initial load.
CE-1.2	Multi-media attachments support (Photos, Videos, PDFs).	P0	Upload PNG, JPG, MP4 up to 20MB with preview.
CE-1.3	Automatic GPS geolocation & interactive Map interface.	P0	Captures lat/long via browser API & Leaflet/Mapbox map pin.
CE-1.4	Multi-lingual UI support (English, Hindi, Santali).	P1	Instant client-side translation toggle.
CE-1.5	Citizen status tracking page via Unique Ticket ID.	P1	Public lookup page rendering live lifecycle stage.

Module 2: AI-Enabled Problem Management & Auto-Routing

Req ID	Requirement Description	Priority	Acceptance Criteria
AI-2.1	Zero-Shot NLP classification of problem description text.	P0	Uses Hugging Face BART-Large-MNLI model with >85% confidence score.
AI-2.2	Automatic Intelligent Routing to relevant Universities.	P0	Maps predicted domain (e.g. Healthcare → Central Univ of Jharkhand) instantly.
AI-2.3	Duplicate Problem Detection.	P1	Flags existing submissions with >80% semantic textual similarity.
AI-2.4	Urgency & Social Impact Scoring.	P2	Algorithmic priority score (1-10) based on severity keywords and votes.

Module 3: University Collaboration & Project Lifecycle

Req ID	Requirement Description	Priority	Acceptance Criteria
UC-3.1	University Queue & Challenge Acceptance/Rejection Interface.	P0	University admins can accept/reject routed issues with notes.
UC-3.2	Team Formation & Faculty Mentor Assignment.	P0	Assign lead student, team members, and faculty advisor.
UC-3.3	Kanban Board Lifecycle Management.	P0	Drag-and-drop state transitions: <i>Assigned</i> → <i>In Progress</i> → <i>Testing</i> → <i>Deployed</i> .
UC-3.4	Milestone Deliverable Tracking.	P1	Track stages: Proposal, Prototype, Field Testing, Deployment.

Module 4: Industry Partnership & CSR Engagement

Req ID	Requirement Description	Priority	Acceptance Criteria
IP-4.1	Project Discovery Catalog for Industry Sponsors.	P0	Searchable directory filtered by domain, stage, funding needed.
IP-4.2	CSR Funding & Resource Pledge Portal.	P1	Record grant pledges, equipment donations, and mentorship allocations.

Module 5: Real-Time Visual Analytics Dashboard

Req ID	Requirement Description	Priority	Acceptance Criteria
AD-5.1	Executive Metric Cards (Total, Active, Resolved, HEIs).	P0	Real-time count cards with auto-refresh every 30 seconds.
AD-5.2	Domain Distribution Pie/Bar Charts.	P0	Interactive Recharts visualization of challenges by sector.
AD-5.3	University Participation & Resolution Leaderboard.	P0	Rankings of top participating universities by projects completed.
AD-5.4	District-Level Heatmap / Geo-analytics.	P1	Spatial plot of challenges across Jharkhand's 24 districts.

7. Non-Functional Requirements (NFRs)

7.1 Performance Requirements

- **Page Load Time:** Client-side initial render <2.5 seconds on 3G networks.
- **API Latency:** FastAPI endpoint response time <300ms for 95th percentile requests.
- **AI Response Time:** Zero-shot classification inference <2.0 seconds per submission.
- **Concurrency:** Support 1,000+ simultaneous active users during peak hours.

7.2 Security & Data Privacy

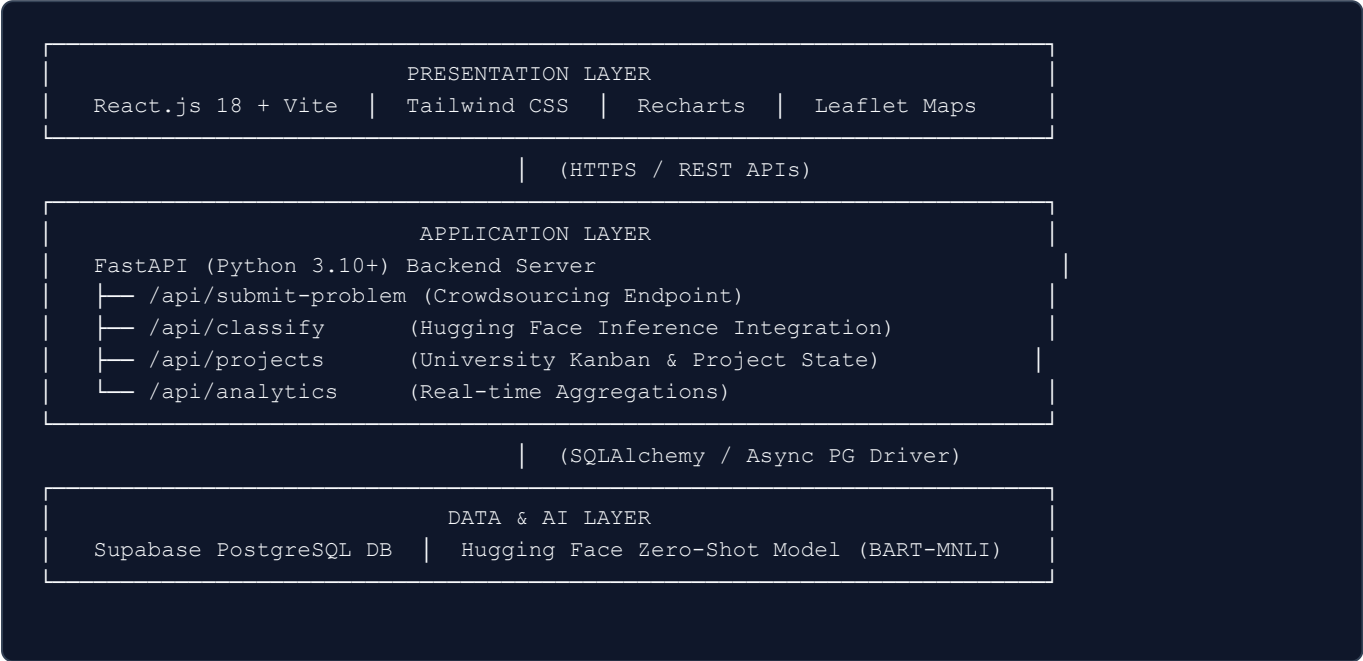
- **Authentication:** Secure JWT (JSON Web Tokens) with 24-hour expiration and refresh tokens.
- **Authorization:** Strict Role-Based Access Control (RBAC) across Citizen, Student, University, Industry, and Admin roles.
- **Data Protection:** TLS 1.3 encryption in transit; AES-256 storage encryption on Supabase PostgreSQL.
- **Compliance:** Strict adherence to the *Digital Personal Data Protection (DPDP) Act 2023* and *Indian IT Act 2000*.

7.3 Scalability & Reliability

- **Stateless Architecture:** FastAPI REST container designed for auto-scaling on Railway/Render.
- **Database Scalability:** Managed Supabase PostgreSQL with automated daily database snapshots.
- **Availability Target:** 99.9% uptime for public crowdsourcing endpoints.

8. Technical Architecture & Tech Stack

8.1 Architectural Layers



8.2 Technology Component Matrix

Component	Technology Chosen	Selection Rationale
Frontend Framework	React 18 (Vite)	Lightning fast client startup, component modularity, ideal for rich dashboards.
Styling Engine	Tailwind CSS	Rapid layout construction, modern aesthetic tokens, responsive utilities.
Backend API	FastAPI (Python)	Asynchronous, automatic Swagger UI docs, native Python AI SDK compatibility.
Database	Supabase (PostgreSQL)	Robust relational integrity, free tier hosting, built-in Auth & Realtime engine.
AI Engine	Hugging Face Inference API	Pre-trained Zero-Shot BART model; eliminates offline ML training overhead.
Mapping	Leaflet.js / OpenStreetMap	Open-source, light footprint, interactive lat/long map picker.

9. Relational Database Schema & Data Models

```
-- 1. Users Table
CREATE TABLE users (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  email VARCHAR(255) UNIQUE NOT NULL,
  full_name VARCHAR(255) NOT NULL,
  role VARCHAR(50) CHECK (role IN ('citizen', 'university_admin', 'student_lead',
'industry', 'government')),
  university_name VARCHAR(255),
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 2. Problems / Challenges Table
CREATE TABLE problems (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  title VARCHAR(255) NOT NULL,
  description TEXT NOT NULL,
  user_category VARCHAR(100),
  ai_predicted_category VARCHAR(100),
  assigned_university VARCHAR(255),
  location VARCHAR(255) NOT NULL,
  latitude DECIMAL(10, 8),
  longitude DECIMAL(11, 8),
  status VARCHAR(50) DEFAULT 'Submitted' CHECK (status IN ('Submitted', 'Assigned', 'In
Progress', 'Testing', 'Deployed', 'Rejected')),
  urgency_score INT DEFAULT 5,
  submitted_by UUID REFERENCES users(id),
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 3. Projects Table
CREATE TABLE projects (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  problem_id UUID REFERENCES problems(id) ON DELETE CASCADE,
  university_name VARCHAR(255) NOT NULL,
  team_lead_name VARCHAR(255) NOT NULL,
  faculty_advisor VARCHAR(255),
  lifecycle_stage VARCHAR(50) DEFAULT 'Proposed',
  proposal_summary TEXT,
  budget_allocated DECIMAL(12,2) DEFAULT 0.00,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 4. Industry Sponsorships Table
CREATE TABLE industry_partnerships (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  project_id UUID REFERENCES projects(id),
  company_name VARCHAR(255) NOT NULL,
  grant_amount DECIMAL(12,2),
  mentorship_lead VARCHAR(255),
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);
```

10. User Interface Layouts & ASCII Wireframes

10.1 Citizen Challenge Submission Page

 SIH 26043 - Societal Innovation Platform

[Language: English ▼]

 Submit a Local Community Challenge

Problem Title *

[Enter a brief title describing the issue...]

Problem Description *

[Detailed explanation of what is happening, who is affected...]

[]

Category Domain (Optional)

[Select Category ▼] (AI will auto-validate: Water, Health, etc.)

Location & Geolocation *

[Enter District/Village] [ Auto-Detect GPS Location]

Upload Photo / Attachment

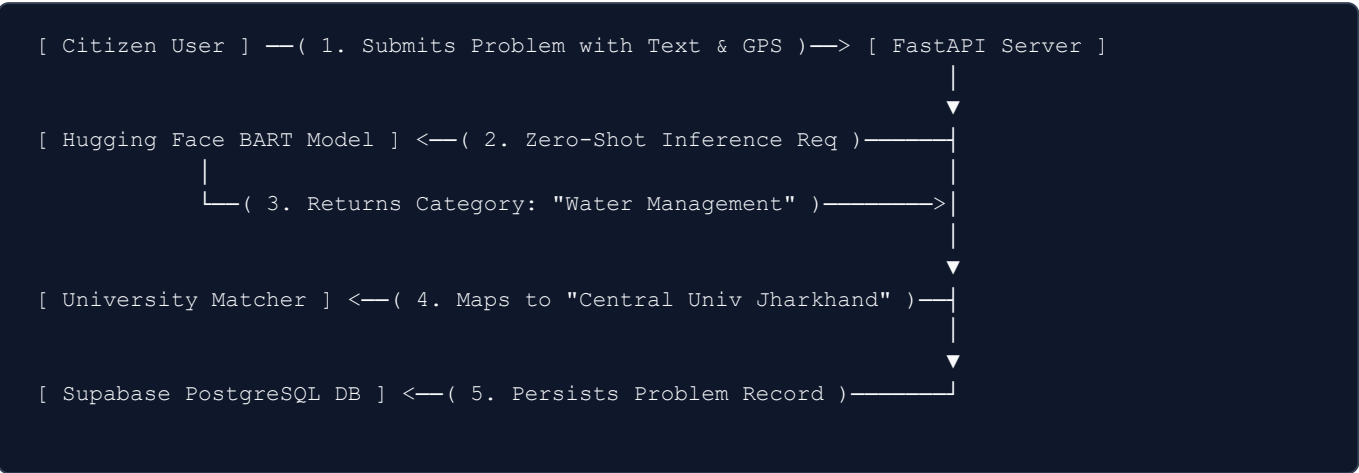
[ Choose File: water_sample.jpg]

[ SUBMIT CHALLENGE TO AI ROUTER]

10.2 University & Government Analytics Dashboard

 Government of Jharkhand - Executive Innovation Dashboard			
Total Problems	Active In-Prog	Deployed Sol.	Participating HEIs
1,247	412	189	48
Domain Distribution (Pie) ■ Water: 35% ■ Health: 25% ■ Agri: 20%		Project Status Lifecycle (Kanban) [Submitted] -> [Assigned] -> [In Progress] -> [Deployed]	

11. Data Flow Diagrams & Sequence Models



12. Security, Privacy & Compliance

- **DPDP Act 2023 Alignment:** Explicit user consent collected prior to processing GPS coordinates and personal contact metadata.
- **API Rate Limiting:** Standard rate limiting of 60 requests/minute per IP to defend against Denial-of-Service (DoS) attacks.
- **Sanitization:** Strict XSS input escaping on description text inputs and SQL parameter binding via ORM.

13. Phased Implementation Roadmap

Phase	Duration	Key Deliverables
Phase 1: MVP Hackathon Prototype	Days 1 - 3	FastAPI auto-categorization endpoint, React submission form, basic visual dashboard, demo script.
Phase 2: Beta Pilot	Months 1 - 3	Supabase Auth integration, multi-language toggles, university project workflow, 10 HEI onboarding.
Phase 3: Statewide Launch	Months 4 - 6	50+ HEIs onboarded, industry CSR portal, offline mobile PWA, automated SMS notifications.

14. Key Performance Indicators (KPIs)

- **Average Triage Time:** Reduction in challenge assignment time from 14 days to <5 seconds.
- **University Participation:** At least 80% of registered HEIs actively working on assigned challenges.
- **Student Engagement:** 1,000+ engineering and research students engaged in real-world NEP 2020 projects.

15. Appendix: 5-Minute Internal Hackathon Demo Script

HACKATHON PRESENTATION STRATEGY

Structure your 5-minute presentation as a live narrative demonstrating real-world impact.

1. **0:00 - 0:45 (The Problem & Vision):** "Imagine a farmer in Bokaro facing water contamination. Currently, there is no direct channel to reach technical experts..."
2. **0:45 - 2:00 (Live Citizen Submission):** Open the React web app. Type a challenge: "*High turbidity and arsenic suspected in school drinking water near Chas village.*" Click Submit.
3. **2:00 - 3:15 (The AI Magic):** Highlight the FastAPI response showing instant zero-shot prediction: "**Water Management**" automatically routed to "**IIT (ISM) Dhanbad - Water Research Division**".
4. **3:15 - 4:30 (University & Admin Dashboard):** Switch to the University Admin tab. Accept the problem, move Kanban stage to *In Progress*, assign a student project team, show real-time analytics chart update.
5. **4:30 - 5:00 (Conclusion & NEP 2020 Linkage):** Conclude with policy alignment: "This turns real-world challenges into academic research pipelines under NEP 2020."